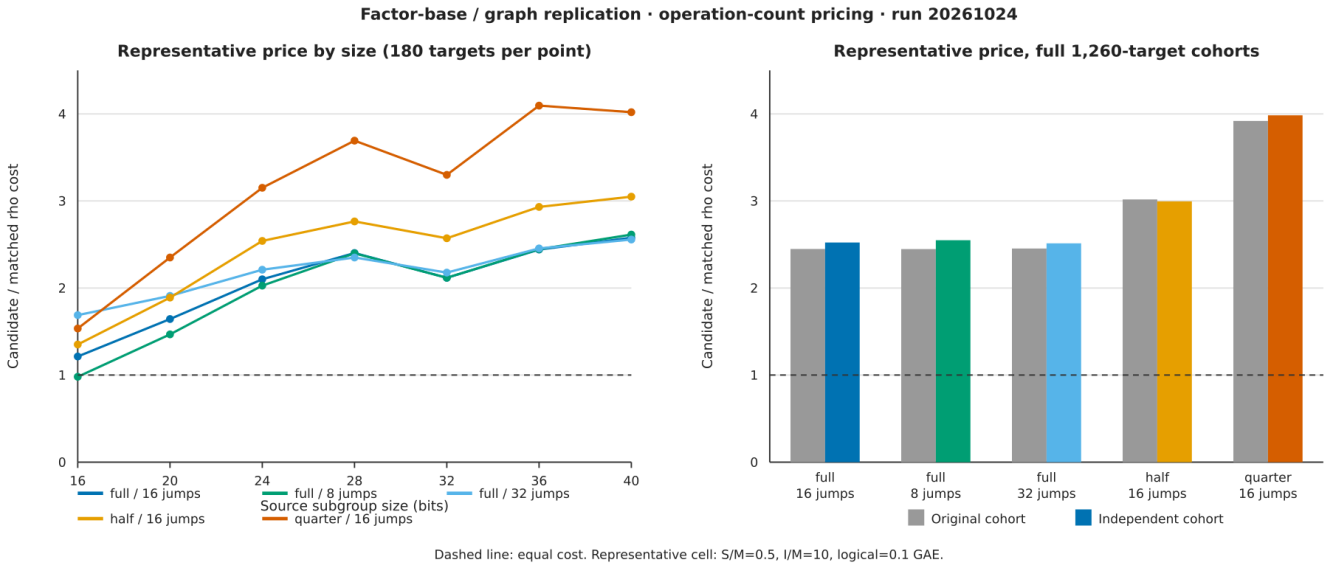
Independent factor-base / graph replication

Run `20261024T2258Z-factor-base-replication` · protocol `factor-base-graph-replication/20261024/v1`

The frozen five-arm protocol ran on a second cohort of 1,260 public known-answer targets. The producer recorded 6,300 candidate solves and 6,300 paired source-negation-rho solves. Every answer was checked by recomputing the scalar multiple on the source curve. Target scalars and algorithm seeds use separate SHA-256 domains. The validator found zero seed, scalar, or target-point overlaps with the original cohort.



Full-cohort operation counts and prices

The representative cell is $S/M = 0.5$, $I/M = 10$, and 0.1 source-addition equivalents per logical operation. Costs are sums over 1,260 targets. The band is the minimum and maximum aggregate ratio over all 36 cells.

Arm	Candidate GAE	Rho GAE	Ratio	36-cell band	W/L/T
Full, 16 jumps	2,661,363,462	1,054,951,912	2.522734	1.386374–13.514033	0/36/0
Full, 32 jumps	2,651,510,986	1,054,951,912	2.513395	1.383840–13.440233	0/36/0
Full, 8 jumps	2,689,869,404	1,054,951,912	2.549755	1.401397–13.639329	0/36/0
Half, 16 jumps	3,159,731,682	1,054,951,912	2.995143	1.715544–16.504039	0/36/0
Quarter, 16 jumps	4,203,242,635	1,054,951,912	3.984298	2.388390–22.870499	0/36/0

One size-only slice has mixed results: full base with eight jumps at 16 bits has 23 winning and 13 losing cells, representative ratio 0.980993, and band 0.790906–2.450390. Every other arm/size slice has zero winning cells.

Original and replication cohorts

Arm	Original	Replication	Ratio delta	Old divisors	New divisors	Delta
Full, 16	2.448830	2.522734	+0.073904	219,581,773	219,259,871	-321,902
Full, 32	2.453278	2.513395	+0.060117	219,864,588	218,282,418	-1,582,170
Full, 8	2.447173	2.549755	+0.102582	220,092,171	221,972,970	+1,880,799
Half, 16	3.017732	2.995143	-0.022589	282,983,105	272,004,613	-10,978,492
Quarter, 16	3.919193	3.984298	+0.065105	392,583,830	387,233,280	-5,350,550

These are two deterministic cohorts with one algorithm seed per target. No sampling interval is assigned; the table records cohort differences without a significance claim.

Validation and scope

The independent native validator recomputed 6,300 target/seed derivations, 12,600 source scalar checks, 40 arm/slice summaries, and all 36 pricing cells per summary. It reports `ok: true`. All 35 raw files have 180 records and exit status 0.

Requirement	Status	Evidence or limit
Independent 1,260-target cohort	Verified complete	Zero seed/scalar/point overlaps
Five arms and paired rho	Verified complete	6,300 candidate plus 6,300 rho records
All costs, wins, and losses	Verified complete	1,440 aggregate price rows
Independent-host replay	Not attempted	No second-host receipt
Isolated wall / external-tool comparison	Not attempted	Counts only; elapsed times diagnostic
Leaderboard promotion	Not proposed	Replay and isolated wall gates absent

The observations apply to the fixed public Scholten models, target derivation, collector, rho implementation, and pricing contract. Machine-readable evidence, checksums, and commands are documented in `DATA_DICTIONARY.md` and `results/manifest.json`.